



Assignment

Spring 2026 2nd Semester

Course Title: Circuit Analysis Lab

Course Code: CMPE-121L

Topic(s): Lab 1 Introduction to Lab Instruments and Components

SUBMITTED TO:

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Post-Lab Questions

1. Define Oscilloscope and write its applications.

An oscilloscope is primarily a voltage-measuring device used to observe and analyze the waveform of electronic signals. It displays a graph of voltage on the vertical axis versus time on the horizontal axis.

Its applications include determining which channels have activity, scaling input signals for display, and measuring the rise and fall time of waves to check circuit performance. It is also used to calculate current and power by measuring voltage and applying Ohm's Law or the Power Law.

2. What is Peak to Peak Voltage?

Peak-to-peak voltage (V_{p-p}) is the total vertical voltage of a signal measured from its maximum positive peak to its minimum negative peak. It is usually twice the value of a single peak voltage (V_p).

3. What is RMS value?

The RMS (root-mean-square) voltage, V_{RMS} , is the value used to calculate the power of an AC signal. It represents the effective DC equivalent of an alternating current signal.

4. Write the RMS values of Sine, Square and Triangular wave?

The RMS value depends on the specific waveform shape and its peak voltage (V_p):

Sine Wave: $V_{rms} = \frac{V_p}{\sqrt{2}} \approx 0.707 \times V_p$

Square Wave: $V_{rms} = V_p$

Triangular Wave: $V_{rms} = \frac{V_p}{\sqrt{3}} \approx 0.577 \times V_p$

5. Explain DC offset?

DC offset is a constant voltage added to a signal that shifts the entire waveform vertically relative to the zero-volt reference. While the default setting is typically 0.0 V, adjusting the offset moves the "center" of the wave up or down on the display.